Learning-Value-Based Credit Assignment for Long-Horizon LLM Agents

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1 Abstract

Long-horizon LLM agents receive weak supervision about which actions cause eventual task success. GRPO broadcasts a trajectory-level advantage across the rollout, while GiGPO introduces step-level credit by grouping repeated anchor states. [4, 3] GACA further shows that credit-assignment granularity can be adapted per state, using policy uncertainty measured by action negative log-likelihood (NLL) to determine how much weight to place on step-level versus episode-level credit. [1] This raises a more fundamental question: *does uncertainty identify the decisions for which additional learning produces the greatest improvement?* I propose a learning-value signal that estimates the expected performance improvement resulting from learning from individual decisions and uses it to adapt RL credit allocation. The initial study will focus on rule-based and statistical estimators, preserving the critic-free structure of GRPO and GACA, before considering learned meta-estimators.

2 Problem and Motivation

The research question is: **Can estimating the learning value of individual decisions improve credit assignment and final task performance in long-horizon RL for LLM agents?**

GRPO provides a critic-free group-relative baseline but assigns the same trajectory-level advantage across all actions. [4] GiGPO addresses this by constructing step-level advantages from trajectories that visit the same anchor state, while retaining an episode-level signal. [3] GACA extends this framework by adapting the mixture between the two levels according to state-dependent policy uncertainty. [1]

GACA establishes an important principle: **credit-assignment granularity should depend on the decision being learned**. Its criticality signal, however, is based on uncertainty rather than learning value. High uncertainty may indicate an important decision, but uncertainty does not directly measure how much future behavior could improve from learning at that decision.

This work instead asks whether the usefulness of a decision can be estimated from the learning process itself. This is inspired by work that reweights training examples according to their effect on a validation objective [5] and by meta-gradient RL, which adapts aspects of the learning process according to subsequent performance. [6] Unlike those approaches, the initial method will not introduce a learned meta-model.

3 Proposed Method

For a trajectory $\tau = (s_1, a_1, \ldots, s_T, a_T)$, define the net learning value of decision t as

$$LV_t = \mathbb{E}\left[\mathcal{P}(\theta + \Delta\theta_t) - \mathcal{P}(\theta)\right],$$

where $\Delta\theta_t$ is the update induced by learning from (s_t, a_t) and $\mathcal{P}$ measures performance on held-out tasks. The quantity represents *net* learning value: an update may be large without being beneficial.

Because directly measuring LV_t for every decision is expensive, the initial study will evaluate inexpensive estimators based on statistics already available during RL, such as update magnitude, gradient direction or alignment, historical learning progress, and outcome sensitivity. These estimators will first be evaluated for their ability to predict measured learning improvement.

The estimated value then controls the mixture between episode- and step-level credit:

$$A_t = (1 - \lambda_t)A_t^{\text{episode}} + \lambda_t A_t^{\text{step}}, \qquad \lambda_t = f(\widehat{LV}_t).$$

Thus,

$$\text{GACA: } \lambda_t = f(\text{NLL}_t), \qquad \text{Proposed: } \lambda_t = f(\widehat{LV}_t).$$

The method remains critic-free and introduces no auxiliary value or learning-value network.

4 Experimental Plan

The primary benchmarks will be long-horizon tool-agent environments such as ALFWorld and WebShop, following the settings used by GiGPO and GACA. [3, 1] The primary models will be Qwen2.5-1.5B and Qwen2.5-7B-Instruct to reproduce GACA's experimental configuration and introduce our method as the only substantive change.

All methods will use matched models, task sets, rollout budgets, optimizers, and training schedules:

1. **GRPO:** trajectory-level credit.
2. **GiGPO:** anchor-state step-level credit.
3. **GACA:** NLL-based adaptive credit.
4. **Learning-value credit:** proposed learning-value-based adaptive credit.

The first experiment tests whether inexpensive estimators predict measured learning improvement at individual decisions. The main experiment tests whether using the best estimator for credit allocation improves held-out task success and learning-curve area at matched rollout budgets.

5 Ablations and Milestones

Ablations will replace the learning-value estimator with GACA's NLL signal, use a fixed mixing coefficient, remove the episode-level component, and compare individual learning-value estimators. [1]

Milestones: Weeks 1–2: reproduce GRPO, GiGPO, and GACA. Weeks 3–4: develop and validate rule-based/statistical learning-value estimators. Weeks 5–6: implement learning-value-based credit and matched-budget training. Weeks 7–8: ablations, held-out evaluation, and analysis.

6 Prerequisites

The first prerequisite is a rigorous understanding of GRPO [4] and GACA [1], including their advantage construction, credit-assignment mechanisms, and training procedures. I will then reproduce the GACA experimental setup and verify that the reproduced baseline matches its reported behavior before introducing any modifications.

Next, I will study existing approaches to estimating the learning value of training decisions, focusing on Meta-Gradient RL [6] and LearnAlign [2], as well as related methods for measuring the effect of individual training examples or updates. This will establish the conceptual and experimental foundation for defining and evaluating learning-value estimators.

Once the baseline and relevant literature are understood and reproduced, I will proceed directly to the estimator validation and matched-budget experiments described above.

References

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